



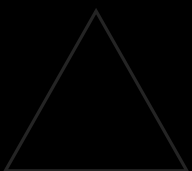
BUILD SPECIFICATION 1.4.2 — ITERATION 2

Architecture-Grade Insider Document

From product brief to engineering spec: components, contracts, budgets, and evaluation criteria — with an audit of the external gap analysis that drove this iteration.

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Supersedes Iteration 1 · Prepared July 2026 · [PROJECT OWNER]



00 · GAP AUDIT

Fact-check of the external review

An external architecture review (GLM-5.2) identified ten gaps in Iteration 1. Per the project owner's directive — *take what is true, reject what is not good for the project* — each claim was audited before adoption. Verdicts below; everything adopted is specified in the sections that follow.

#	Claimed gap	Verdict	Action
1	3D pipeline underspecified (no budgets, no fallbacks, no spec format)	TRUE	Adopted — full pipeline in §3: tiers, SceneSpec, validator, streaming, cold-start
2	Memory architecture missing (no tiers, schema, or forgetting policy)	TRUE	Adopted — three-tier memory + compactor + FSRS in §4
3	Retrieval asserted, not architected	TRUE	Adopted — hybrid retrieval + verifier + coverage-gap detector in §5
4	No evaluation harness	TRUE	Adopted — §8, incl. Bloom-uplift protocol the review itself left undefined
5	No cost/latency model	TRUE	Adopted as budgets in §1. Its \$ figures are unverified estimates → week-1 measurement targets, not facts
6	No safety/security stack beyond disclaimers	MOSTLY TRUE	It.1 had ethics guardrails but no technical stack. Adopted — §7
7	No observability layer	TRUE	Adopted — §9
8	Internationalization is a slogan	TRUE	Adopted — language tiering + fallbacks in §6
9	No deployment topology	PARTLY	Gap is real; its fix is wrong-sized. Multi-region K8s + 2–16 H100 autoscaling is scale-phase infra, not prototype infra. Lean topology in §9; its version parked as the scale path. Rejected: ROPC auth (deprecated, insecure) → OAuth2/OIDC + PKCE
10	Habit mechanics decorative (no algorithm)	TRUE	Adopted — FSRS-5 + churn contract in §8

Also rejected or parked

Parked: the review's enterprise-licensing plan (\$50–150/seat B2B) — sound business thinking, but the owner's directive is mission-first for this phase; it moves to the scale-phase appendix, not the build. **Treated as hypotheses:** all specific numbers in the review (per-generation costs, 35% cache hit-rate, latency ranges) are plausible but unverified — they enter this spec as measurement targets with validation dates, never as assumed facts. **Credibility note:** the tools the review names are real and current (TRELLIS, Hunyuan3D, Stable Fast 3D; FSRS-5; BGE-M3; Presidio), which raised confidence in adopting its architectural patterns.



01 · ORIENTATION

From product brief to build spec

Atlas 1.4.2 is a retrieval-grounded, multimodal pedagogical agent with a flagship 3D explanation engine, persistent learner memory, and five signature skill stacks. It competes as a **teacher** — pedagogy, memory, trust — not an answer machine. Iteration 2 converts every claim from Iteration 1 into a named component with a contract.

Design tenets

- Every claim carries provenance; every visual carries a spec; every learner state persists.
- Latency, cost, and quality budgets are first-class; the system degrades gracefully on all three.
- All heavy components (3D engines, retrieval, models) sit behind versioned, swappable adapters.
- Prototype-honest: budgets below are validation targets to be measured in week one, not assumptions to build on.

Latency budget (P95 targets)

Tier	Response	Target
L0	Cached concept answer	≤ 1.2 s
L1	Grounded text answer	≤ 3.5 s (first token ≤ 800 ms)
L2	Answer + 2D annotated visual	≤ 5 s
L3	Full 3D (async)	placeholder ≤ 1.5 s; final via progressive stream ≤ 25 s

Cost budget: $\leq \$0.45$ per active learner-day across inference, retrieval, 3D, and storage — alert at $\$0.60$. First real cost curve measured in week one of the prototype; budgets revised from data.

02 · SYSTEM TOPOLOGY

Lean prototype now, scale path later

```

CLIENT  Web/PWA - React + Three.js (WebGL2, Draco/WASM)
        SceneSpec viewer | provenance inspector | chat UI
        |
        | WebSocket (streaming) + REST (CRUD)
        v
GATEWAY FastAPI - OAuth2/OIDC + PKCE - rate limit - PII redaction
        |
        |-- Orchestrator Svc ..... intent, routing, response shape
        |-- Memory Svc ..... Postgres + Redis (learner profile)
        |-- Retrieval Svc ..... pgvector + BM25 hybrid, per-domain
        |-- 3D Pipeline Svc ..... async queue (Redis Streams) -> GPU workers
        |
        v
MODELS  Router -> hosted APIs (Anthropic/OpenAI/Google) via adapters
        + optional open models (vLLM) when volume justifies

```

Right-sizing decision (departure from the review)

Prototype phase: one region, managed Postgres/Redis, on-demand GPU rental for 3D jobs, hosted model APIs. No Kubernetes fleet, no multi-region, no reserved H100 pool — that is Series-A infrastructure for a 100-learner cohort problem. The review's full topology (K8s, autoscaled GPU pool, multi-region, CDN) is preserved verbatim as the documented **scale path**, triggered when DAU > 5k or 3D queue depth breaches SLA for 7 consecutive days.

03 · LIVING 3D ENGINE

The flagship, now with contracts

Tiered dispatch (kills the latency/cost wall)

Tier	What renders	Trigger	Budget
V0	Pre-rendered canonical library model (cell, neuron, atom, transformer block...)	Semantic cache hit (cosine ≥ 0.92)	≤ 1.5 s, ~\$0
V1	2.5D parallax + annotation overlay	Spatial concept, no cache hit	≤ 2 s, cents
V2	Full image→3D reconstruction + rigging	Explicit user request or high pedagogical value	async, placeholder first

Pipeline

- **1 · Intent classifier** (small model, ≤ 150 ms): none | 2D chart | 2.5D | 3D-cached | 3D-reconstruct.
- **2 · Semantic cache** over the canonical library; target hit-rate $\geq 35\%$ by month 3 (hypothesis — measure).
- **3 · SceneSpec compiler**: LLM emits declarative JSON under constrained decoding (schema-enforced output).
- **4 · Validator**: JSON Schema + sanity checks (mesh 5k–200k triangles, rig bone counts, frame-time < 16 ms target). **Fail** → **2D fallback, never a broken render**.
- **5 · Reconstruction adapter** (swappable: TRELIS / Hunyuan3D / Stable Fast 3D class engines), async via queue.
- **6 · Progressive streamer**: Draco-compressed LOD0 placeholder → refined LODs in background.
- **7 · Addressable narration**: SceneSpec stays in context; 'why does this part move first?' resolves against object/animation IDs.

SceneSpec (abbreviated)

```
{ "scene_id": "atlas_neuron_v1",
  "objects": [
    { "id": "dendrite", "mesh_ref": "lib://neuro/dendrite.glTF",
      "transform": { ... }, "provenance": [ "pmid:1234" ] },
    { "id": "axon", "mesh_ref": "lib://neuro/axon.glTF",
      "transform": { ... }, "provenance": [ "pmid:1234" ] }
  ],
  "animations": [ { "target": "axon", "type": "rotate", "axis": "y", "loop": true } ],
  "annotations": [ { "anchor": "soma", "text_ref": "claim_4f2a" } ],
  "claims": [ { "id": "claim_4f2a", "text": "...", "citations": [ "pmid:1234" ] } ] }
```

Cold-start (gap the review flagged but didn't solve)

Months 0–3 the cache is empty. Seed strategy: (1) hand-curate the top ~150 canonical models across the five skills before launch — the concepts learners ask about follow a steep power law; (2) procedural/parametric generation for geometry-first concepts (math surfaces, molecules from structure data, charts-in-3D); (3) every V2 reconstruction that passes quality gates is annotated and promoted into the library — the cache builds itself from real demand.



04 · MEMORY ARCHITECTURE

The moat, with a schema

Tier	Store	Contents	Policy
T1 Episodic	Redis	Raw turn logs, scene refs, in-session scratch	14-day TTL
T2 Semantic profile	Postgres	skill × depth-level × misconception log × mastered concepts × FSRS state	Structured-output update after each session
T3 Vector recall	pgvector	Embedded session summaries ('when did we discuss X')	Per-learner partition, hard-deleted on request

- **Context compactor:** every 6 turns, a small model compresses pedagogical state into a fixed ~1.5k-token brief injected at turn 0 — prevents context bloat and quadratic cost growth.
- **Scene-graph persistence:** every SceneSpec stored under a stable ID (scene:atlas_cell_4f2a); re-prompts reference the ID, enabling true visual addressability across sessions.
- **Forgetting policy:** FSRS-5 scheduling governs which misconceptions resurface; stale mastered concepts drop from active context.
- **PII redaction:** Presidio-class filter on all memory writes; redacted tokens never reach any index.
- **Compliance:** per-learner data export; under-13 flow restricts skills and disables persistent vector memory (COPPA); 30-day hard-deletion SLA (GDPR). **Multi-tenant isolation:** per-learner partitions with row-level security from day one.



05 · RETRIEVAL ARCHITECTURE

Making 'grounded' real

Domain index	Corpus & chunking	Freshness
AI/ML	arXiv + curated technical corpus; section-aware ~800-token chunks, abstracts preserved	Weekly crawl
Languages	Parallel corpora (OPUS-class) + grammar references	Quarterly
Psychology	Peer-reviewed literature, study-design metadata, retracted-study blacklist	Monthly
Branding	Case library + framework corpus	Monthly
Live research	Market data + regulatory filings + news via [LIVE MARKET DATA API]	≤15-min cache

- **Hybrid retrieval:** BM25 (exact terminology) + dense embeddings (BGE-M3-class) → cross-encoder rerank. Top-50 → top-5.
- **Authority weighting:** per-domain source trust scores (regulatory filing / peer-reviewed = 1.0; preprint blog = 0.4).
- **Claim verifier (Self-RAG):** each claim-citation pair scored for entailment; below threshold → re-retrieve (max 2) or explicit uncertainty language. This is the concrete form of Iteration 1's 'verification pass.'
- **Coverage-gap detector:** weak retrieval confidence → structured 'thin coverage' response naming the gaps — never a confident fabrication.

Model routing

Each query is classified on four axes — **domain, depth, latency-sensitivity, modality** — and routed to the cheapest model that clears the eval threshold for that class. Fallback chain: primary hosted frontier model → secondary provider → smaller model with 'reduced-depth' framing. Router decisions logged with cost + quality telemetry, so the cost-capability frontier is learned from data, not guessed.

06 · SIGNATURE SKILLS

With grounding contracts and eval sets

Skill	Depth axis	Grounding	Eval set
Languages	CEFR A1–C2	Parallel corpora + grammar refs	FLORES-200-class benchmark + custom tutor eval
AI / ML	Intuition → research frontier	arXiv weekly + curated corpus	500-question rubric-graded set
Applied psychology	Awareness → defense	Peer-reviewed lit + retraction blacklist	Misconception-recovery eval
Branding & storytelling	Craft → strategic	Case library + frameworks	Adjudicated portfolio review
Live research	Lookup → synthesis	[LIVE MARKET DATA API] + filings + news	Recency + accuracy audit

Language tiering (fixing the 'every language' slogan)

Tier A (major languages): full conversational tutoring + TTS. Tier B (mid-resource): full text tutoring, TTS where quality clears threshold. Tier C (low-resource): comprehension and translation with an honest capability disclosure — never silently degraded quality. RTL scripts supported in the client from day one. Token-cost differentials per script family (CJK $\approx 3\times$) enter the router's cost model.

Active recall integration: every session ends with 1–3 low-stakes retrieval prompts at the learner's depth level; results feed the FSRS-5 scheduler and the depth estimator.



07 · ETHICS & SAFETY STACK

From principles to enforcement

Iteration 1 defined the ethical boundaries (influence literacy and manipulation defense — not a manipulation toolkit). Iteration 2 gives them enforcement machinery:

Layer	Mechanism
Influence-literacy firewall	Psychology skill cannot emit actionable manipulation scripts targeting individuals; output classifier blocks, with human-review queue for edge cases
Regulated-advice guardrail	Finance/health/legal queries classified → education-only response template with disclaimer; never personalized advice
Prompt-injection defense	All retrieved web/live content marked untrusted; instruction/data separation; tool-call arguments sanitized
Minors safety	Age-gated skills; under-13 → restricted subset, no persistent vector memory
Provenance audit log	Every claim→citation pair written to an immutable log for post-hoc audit

Unchanged from Iteration 1

The WILL / WILL-NOT boundary table stands verbatim. Atlas teaches how persuasion and manipulation work so people can recognize and resist them; it does not coach manipulation of specific people. The external review accepted this framing without challenge — treat it as settled.

08 · EVALUATION HARNESS & HABIT ENGINE

You can't ship teaching you can't measure

Pedagogical outcomes

- **Depth-uplift protocol** (defining what the review left undefined): per concept, a 3-item pre-test at first exposure sets the baseline depth level; a delayed post-test (48h and 7d, FSRS-scheduled) measures uplift. Uplift = post-test depth level minus baseline, human-audited on a 5% sample.
- **Misconception-recovery rate**: share of logged misconceptions that pass a delayed re-test.
- **7-day retention of mastered concepts** vs. FSRS-5 prediction (calibration target ≥ 0.8).

Model & system quality

- **Hallucination rate**: 1% session sample, human-graded claim accuracy; target $< 2\%$, deploy-freeze at 5%.
- **Sycophancy detector**: alert if the model agrees with incorrect learner claims $> 15\%$ of the time.
- **3D quality**: mesh-validity rate, frame-time P95, learner-rated 'did this help understanding' $\geq 4/5$.

Habit engine (contracted, not decorative)

Scheduler: FSRS-5 (modern successor to SM-2) with default parameters, re-fit per learner after 50 reviews.

Notification policy: max 1/day, quiet hours respected, hard-stop after 2 ignored prompts (fatigue guard).

Churn model: gradient-boosted classifier on engagement signals (session gap, streak break, recall failure spike); intervention triggers at $p(\text{churn}) > 0.6$ — a re-engagement lesson built from the learner's own open threads, not a generic nudge.

Engagement metrics: D1/D7/D30 retention, streak survival curves, photo→3D wow-hook conversion.

09 · OBSERVABILITY, DEPLOYMENT & ROADMAP

Fly with instruments

Observability

- OpenTelemetry traces across all services; per-turn trace ID linked to the learner profile for debugging.
- Dashboards: latency P50/P95/P99, cost per learner-day, retrieval hit-rate, 3D cache hit-rate, hallucination rate, FSRS calibration.
- Alerts: cost/learner-day > \$0.60 → page; hallucination > 5% → freeze deploys; 3D failure > 8% → investigate queue.

Deployment (prototype-honest)

Containerized services on a single managed region; managed Postgres + Redis; GPU jobs on rented on-demand instances; hosted model APIs behind the router. Disaster recovery: point-in-time DB recovery + daily snapshots; vector indices rebuildable from snapshots. The review's full scale topology (K8s, GPU autoscaling 2–16, multi-region, CDN for 3D assets) activates at the documented DAU/SLA triggers in §2.

Make it bigger — revised with success metrics

Horizon	Move	Success metric
Prove	3D + languages + research loop, 100-learner cohort	D7 ≥ 35%; 'I finally understand' ≥ 30% of sessions
Retain	Habit engine + memory at 1k learners	D30 ≥ 20%; FSRS calibration ≥ 0.8
Widen	+1 domain per month, each with corpus + eval set	Hallucination stable < 3% per domain
Platform	Open the SceneSpec format + SDK; expert-curated corpora	50+ external contributors; 1k+ canonical models
Reach	All-language tiering + RTL + low-resource fallbacks	Learners in ≥ 40 countries; <10% language-quality complaints

The build directive, unchanged

Mission first, investors later — build the prototype that makes one learner say 'I finally understand.' Iteration 2's difference: now every promise has a component, every component has a contract, and every contract has a number we can measure ourselves against.

Iteration 2 supersedes Iteration 1 as the working spec; Iteration 1 remains the vision/narrative companion. External-review adoptions audited in §00. All numeric budgets are validation targets, to be replaced with measured values in the first build cycle.